Machine Learning Exercise 1

Problem 1 *In this exercise, you will implement linear regression with one variable to predict profits for a food truck. Suppose you are the CEO of a restaurant franchise and are considering different cities for opening a new outlet. The chain already has trucks in various cities and you have data for profits and populations from the cities. You would like to use this data to help you select which city to expand to next.*

The file `ex1data1.txt` *contains the dataset for our linear regression problem. The first column is the population of a city and the second column is the profit of a food truck in that city. A negative value for profit indicates a loss.*

i) *Plot the data.*

ii) *Define the cost function.*

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2,$$

with hypothesis function:

$$h_\theta(x) = \theta^T x = \theta_0 + \theta_1 x_1.$$

iii) *Use gradient descent to fit the linear regression parameters θ.*

Recall that the parameters of your model are the θ_j values. These are the values you will adjust to minimize cost $J(\theta)$. One way to do this is to use the batch gradient descent algorithm. In batch gradient descent, each iteration performs the update

$$\theta_j := \theta_j - \alpha\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})x_j^{(i)}$$

You can set the learning rate $\alpha = 0.01$, and initialize θ to zeros. Simultaneously update θ_j for all j. With each step of gradient descent, your parameters θ_j come closer to the optimal values that will achieve the lowest cost $J(\theta)$.

iv) *Check the progress of the gradient descent by checking that $J(\theta)$ is decreasing after each step.*

v) *Plot the fit with the adjusted values of θ and compare to the initial and some intermidiate values of θ.*

vi) *Check if you get the same values of θ using scikit-learn's* `LinearRegression` *or NumPy's* `polyfit` *in Python.*

vii) *Use the final values of θ to make predictions on profits in areas of $35,000$ and $70,000$ people.*

viii) *Plot the cost function $J(\theta)$ over a 2-dimensional grid of θ_0 and θ_1 values.*